

Business Requirements Document – Enrollment System (Consulting-Grade Narrative Version)

1. Executive Summary

The Enrollment System is a comprehensive, web-based admissions platform designed to manage the full lifecycle of school admissions across multiple schools, each with independent configuration. It enables parents to register, submit applications, receive offers, complete enrollments, and track status, while administrators manage selection criteria, lotteries, waiting lists, communications, and compliance requirements. The platform is built for high-volume usage: in one live deployment, over 6,000 applications were handled in a single year, and by the end of year two the system processed more than 6,500 applications in *two days*[1]. The system aims to eliminate manual processing, reduce errors, ensure fairness and transparency in selection (through auditable lottery procedures), enforce documentation compliance, and provide clear, timely communication to all stakeholders. It must support demanding loads (thousands of concurrent submissions) and enable seamless parent and school experiences.

2. Purpose

The purpose of this system is to provide a centralized, reliable, and scalable admissions solution for schools. It supports two distinct stages: an **Application** stage (parents apply to school) and an **Enrollment** stage (accepted students provide additional data)[2]. Each school can configure its own application form, criteria, and capacities, while super-administrators manage global settings and template consistency. By automating data capture, validation, review, selection, and communication, the system ensures every application is processed consistently, accurately, and in accordance with each school's rules and legal obligations. The system enforces submission deadlines, document requirements, and admissions policies (e.g., lottery procedures), thereby improving fairness and transparency. Reliable audit logging and exportable records support regulatory compliance and decision-making.

3. Scope

This BRD covers all functional and non-functional requirements for the end-to-end admissions workflow. It includes **Application Intake, Parent Registration, Application Form entry and submission, Criteria Evaluation and Lottery Selection, Offer/Waiting List Management, Offer Acceptance and Decline flows, Enrollment Form and Data Collection**, as well as all **Administrative Configuration, Communications, and Reporting/Export** features. It addresses both parent-side and administrator-side interfaces. For example, the system must guide parents through a multi-page Application form (collecting child and guardian information, demographic data, documents, and admissions criteria answers), then transition accepted applicants into a separate

Enrollment form (collecting medical, custody, consent, and other details)[2][3]. It also includes admin tools for configuring school settings (such as open/close dates, capacities, questions, and consents), running the selection/lottery process, issuing offers, and generating reports. All stages have associated automated communications (emails to parents) and audit trails. Any referenced laws (e.g. public admissions timelines) must be configurable per school.

4. Stakeholders & User Groups

- **Parents/Guardians** – End users who create accounts, submit one or more child applications, upload required documents, and accept/decline offers.
- **School Administrators** – Users who configure their school’s admissions cycle (dates, capacities, application fields, criteria, communications), review applications, verify documents, run the selection/lottery process, issue offers, manage the waiting list, and oversee enrollments.
- **Super Administrators** – System-level administrators who manage global configuration: onboarding new schools, defining system-wide dropdown values (e.g. list of countries, ethnicities), default templates, and cross-school reporting.
- **Support/IT Teams** – Provide operational support, troubleshooting, and maintenance (e.g. handling high support load for login or upload issues as noted by dozens of parent tickets[4]).
- **Auditors/Compliance Officers** – Review logs and reports to ensure admissions were conducted according to policy.
- **School Leadership** – Use reports and dashboards for decision-making and oversight.

5. Current Challenges & Pain Points

In existing manual or legacy systems, schools face delays and errors due to inconsistent data capture and cumbersome processes. Common issues include:

- **Inconsistent Data Entry and Required Fields:** Schools have different policies about which questions to ask. In practice, some questions that *should* be optional were hard-coded as mandatory (for example, one school’s nationality field is marked required even though policy indicates it should be optional)[5]. The new system must allow each school to turn fields on or off and mark them mandatory, so parents only see relevant questions (stars denote mandatory fields)[6][7]. Any mismatches (e.g. mandatory toggles not working) must be resolved.
- **Document Validation and Upload Issues:** Documents must be securely uploaded and verifiable. In demonstrations, parents were prompted to upload a birth certificate and proof of address[8][9]. However, parents could not preview uploads before submission[8], and generated PDFs were incomplete (missing certain answers like an ASD class selection[10]). The system must enforce allowed file

types, clearly indicate upload requirements, enable preview/download for administrators, and generate correct PDFs for both applications and enrollments.

- **Application Identification and Visibility:** Every submitted application gets a unique number. Currently, parents see that number but it is not visible to administrators until final submission[11]. The lack of a central list of all registered parents (and drafts) can cause confusion (the school cannot see draft applications)[11]. The design should clarify when and how admins can view applicant lists.
- **Export and Pagination Limitations:** Administrators often work with exported data in spreadsheets. The current system can only export ~70 rows at a time, splitting data across multiple files[12]. Large schools have hundreds of applications, so the export feature must support unlimited rows and flatten data into one comprehensive file. Likewise, in the on-screen lottery view, pagination was limited to 50 entries, making it hard to find numbers on other pages[13]. Sorting and searching in these views must function reliably.
- **User Interface Inconsistencies:** Several UI issues were noted during demos. For example, clicking “Next” on a form page did not scroll the window to top, causing parents to miss the beginning of the next section[14]. The consent form for data use is shown halfway through the application (rather than at the end), causing a confusing flow[15]. Also, in the account setup, parents could not upload a signature image due to a bug[16]. The configuration interface is fragmented (some field toggles appear in one tab, related settings in another)[17], which needs clean-up.
- **Session Timeout/Logout:** Administrators reviewing applications experienced a “soft logout” bug: the interface froze or hid options after idling ~10 minutes[18]. This disrupted review work. Session handling must be robust so admins don’t lose progress when validating data.
- **Audit Trail Gaps:** The system provides an “application trail” showing timestamps of status changes[19]. However, any missing logs or unclear terminology (e.g. what exactly triggers a “submitted” vs “applied” status) can cause confusion. All actions (submission, offer sending, acceptance, decline, edits) must be unambiguously recorded.

These issues will be addressed in the new system design.

6. End-to-End Process Overview

1. **Parent Account Registration:** A parent visits the school’s portal, clicks the enrollment link, and creates an account (entering name, email, and phone)[20]. The system sends a verification code via email and an OTP via SMS for phone confirmation[20]. Once verified, the parent sets a password and (optionally)

uploads a signature image (note: the current system's signature upload is broken[16], but the design must allow either drawing or image upload).

2. **Application Start (Draft):** After login, the parent sees the school's welcome page (customizable by the school, including school-provided text and links to policies[21]). A pop-up reminds the parent to gather required documents (e.g. birth certificate, ID, proof of address) before beginning[22]. The parent clicks "Add Child" (the plus icon) to start an application. At this point the application is saved as a "Draft" and given an application number. Until the parent submits the form, the school **cannot see or access this draft application**[11]. (Administrators will only see submitted applications.)
3. **Application Form Completion:** The parent proceeds through a multi-page form. Fields appear or hide based on earlier answers and school-configured settings. Mandatory fields are marked with a red star. The pages typically include:
4. **Student Details:** Legal name (as on birth certificate), date of birth (the system enforces age rules – e.g. children must be ≥ 11.9 years old to apply for 1st year[6]), PPS number, religion, gender, country of origin, nationality, etc. (Some schools choose which of these to ask. In one example, the school made religion mandatory[6], while gender, origin, nationality were optional toggles.)
5. **Document Upload (Application Stage):** If configured by the school, parents are asked to upload a copy of the birth certificate at this point[8]. The system should accept the file and store it with metadata (but note: in testing the preview function did not display the uploaded image to the parent, an issue to fix[8]).
6. **Child's Address:** The form then asks for the child's home address and Eircode (postal code). These are required to proceed[23]. (The parent cannot navigate past this page without entering address.)
7. **Guardian 1 Details:** Next the form collects primary guardian (parent) information: name, relationship (e.g. "Mother"), mobile phone (not verified again by OTP), email (also not re-verified), and address (either same as child or entered manually). For example, the demo admin entered himself as "Mother" of the child with same address[24].
8. **Guardian 2 (Optional):** The parent may add a second guardian by clicking the plus sign. The form then shows fields for the second guardian's name, relationship (e.g. "Father"), phone, and email. In the demo, the second guardian's contact details were filled (again no OTP needed for these) and address could be copied from Guardian 1[25].
9. **Contact Preferences:** A page asks which guardian(s) should receive school communications (email, SMS, letters). Parents can select one or both for each channel[26]. (Behind the scenes, the system is intended to send notifications accordingly, though one demo found only the primary contact was receiving emails even when "both" was selected[27].)

10. **Review & Consent (Application Stage):** The parent is then shown a review of entered information (the demo indicated a “sharing of data” consent was shown mid-form, though this is slated to move to the end)[15]. The parent must agree to a data usage statement (“We are using your data for...”) before proceeding. (Eventually, they must also agree to the school’s Code of Behaviour and anti-bullying policy at the very end of the application, after all data is entered.)
11. **Primary School Details:** Next the form asks for the current primary school: name, address, attendance dates. The demo showed a flaw: the prompt asked for address but no address field was present (parents had to type it in manually)[28]. This must be corrected so the primary school’s name and address are captured in structured fields.
12. **Parent-Defined Criteria Questions:** The core criteria page appears last. The school can configure a series of yes/no questions for admissions priority (e.g. “Child has a sibling at this school?”, “Child attends a feeder school?”, etc.). The parent answers Yes or No to each. If a question has follow-up requirements, additional fields appear only when the parent answers Yes. For example, when the parent answered Yes to “sibling at school”, the form displayed fields to enter the sibling’s name and class[29]. Each criterion is configured by the school in priority order (e.g. sibling = highest priority). The system **does not compute numerical scores** for each applicant; instead, it records the yes/no answers and any additional info. (The demo stressed that the first criteria answered “Yes” essentially means top priority, but no numeric score is shown[30].)
13. **Special Education Question:** The form includes any specialized queries, such as autism support needs. In the example, there was a checkbox “Is child applying for a place in the ASD unit?” followed by “If no place, apply to mainstream?” parents. A demo issue was noted where if the parent said Yes, the follow-up “Mainstream offer?” automatically defaulted to No – which is incorrect; it should have been blank until answered[31]. This must be fixed so parents explicitly answer the second part.
14. **Conditional Document Uploads:** For certain criteria, the school might require proof documents. In the ASD scenario, the parent answered Yes and then the form *required* uploading an NCSC support letter and a utility bill for address[9]. The parent uploaded files in the form and signed a consent. The system should support these conditional document requests (the admin can configure which questions trigger additional uploads[32]).

After all pages are complete, the parent is presented with the final Code of Behaviour and anti-bullying policies, and must sign and consent to each[33]. On submission, the system records the application with timestamp, sends a confirmation email to the parent with a PDF of the application, and finalizes the application status. In the demo, the confirmation email arrived immediately upon submission[34]. The attached PDF was noted to be poorly formatted and incomplete (missing the ASD answers)[34] – this is unacceptable and must be corrected so that the PDF accurately reflects every filled field. After submission, parents may **cancel** their application (mark it cancelled) or **delete** it. (In the demo, “Cancel” changed the status to Cancelled[35]; the effect of “Delete” was unclear and

requires clarification.) Importantly, once an application is submitted and signed, the parent **cannot edit it**, only cancel it[36] (to ensure integrity of the application).

7. Application Intake Management

Each school's intake is defined by grade/class. The system allows administrators to set the number of available places per grade for each admissions cycle[37]. These capacities (intakes) can be updated as needed before the selection process begins. The school also sets open and close dates/times for the application window. These dates appear on the parent dashboard (the demo showed the open/close dates auto-displayed)[21][38]. Once the close date passes, late applications (submitted after the deadline) are automatically flagged as "late" and excluded from the standard selection process[39]. (They may be placed at the bottom of a waiting list or handled per policy.)

When applications exceed capacity, the system triggers the selection process. It filters all submitted applications through the configured criteria in priority order. For each criterion (highest priority first), any applicant answering Yes is selected until no more seats remain at that level[40]. If the number of qualifiers at a given criterion exceeds remaining seats, the system invokes a lottery among those applicants[41]. Finally, any applicants still without a place (because seats are filled) are placed on the waiting list (see sections 14 and 15). This automated workflow eliminates manual seat assignment. Administrators have full visibility into this process via the admin interface.

8. Parent Application Form Requirements

The parent application interface must guide the user through the multi-step form with clarity. Key requirements and behaviors include:

- **Dynamic Fields:** Only questions selected by the school's configuration are shown. Mandatory fields (as indicated by a star) must be completed before moving on[6]. Optional fields do not block progress. For example, in the demo any field *below* a certain point was optional and could be toggled off by the school[5]. The system should enforce these rules strictly (e.g. parents could not advance past the address page without providing the child's address[23]).
- **Pop-up Instructions:** Upon starting a new application (clicking "Begin"), the parent sees a modal reminding them to have key documents ready (birth certificate, ID numbers, utility bill, etc.)[22]. This helps reduce incomplete submissions.
- **Field Validation:** Certain fields have special validation. For instance, the child's date of birth must satisfy the minimum age requirement (e.g. at least 11.9 years old by admission date)[6]. The PPS number is validated for format. Any validation errors should be clearly shown. In the demo, entering an age below the cutoff prevented proceeding[6].

- **Document Uploads:** The form supports multiple upload questions at both Application and Enrollment stages. For application stage, the school can require specific documents (e.g. birth certificate, NCSC letters). Parents upload files directly in the form. All uploads should be stored securely with metadata (which question and application they belong to). Parents should be able to preview uploaded files before submission (in the demo, the upload appeared but could not be viewed[8]; this must be fixed). The system must also prevent incorrect uploads (mismatched file to question).
- **Progress and Navigation:** Parents navigate through the form pages via Next/Back buttons. The interface should automatically scroll to the top of each new page (the demo had a bug where Next left the viewport at the bottom[14]). Parents can review their entered data at any point (especially on the final page). An explicit review screen before submission is encouraged to catch errors.
- **Saving and Drafts:** Applications can be saved as draft (as soon as the child is “added”). Parents should be able to log out and resume later. Administrators must not see draft applications until submission (for data privacy)[11]. Parents may delete drafts freely.
- **No Double Application:** Each parent must only have one active application per child per school. The system should prevent duplicate submissions for the same student in the same cycle.

9. Document Upload Management

The system must manage all required documentation upload seamlessly:

- **Multiple Document Types:** Apart from default ones (birth certificate, proof of address), administrators can configure *Document Requests* for any stage[32]. For example, a school might require proof of custody or medical forms if a related question is answered yes[9][32]. These appear as additional upload fields in the form.
- **Mandatory Uploads:** For each required document, parents must provide the file before continuing. The system should clearly indicate which uploads are required at each stage (e.g. “Upload utility bill” or “Upload doctor’s letter”).
- **Storage and Security:** All uploaded files are stored in a secure repository. Each file is tagged with application ID, question identifier, and stage (Application vs. Enrollment). Access is restricted by role (only relevant admins can view).
- **Viewing and Preview:** Administrators must be able to preview or download any uploaded document from the application’s detail view[42]. In the demo, clicking to view the birth certificate showed the uploaded image at its original resolution[42]. This preview feature must be robust for PDF, image, and other formats.

- **Batch Operations:** Admins may need to download multiple documents (e.g. all birth certificates). The system should allow bulk downloading (e.g. as a zip file).
- **Conflict Prevention:** The system should detect obvious mismatches (e.g. uploaded birth certificate does not match the entered child's name). Basic checks can alert the admin.
- **Generation of Enrollment Document:** After Enrollment form submission, a final Enrollment PDF is generated (combining all forms and documents). This PDF must be complete and legible. (In the demo, the "application submission" email included a PDF copy, but the Enrollment form's PDF was reported as "horrible" and actually contained the wrong data[43]. This is a high-priority fix.)

10. Offer Management

Administrators need clear, controlled workflows to issue offers:

- **Selection Trigger:** After the close date has passed and all applications are entered, the admin triggers the selection process. Until then, offer-related actions are disabled[44][45].
- **Automated Offer Emails:** Once selection (criteria filtering and lottery) is complete, the system marks selected applicants as "Offered." Admins click "Send Offers," which generates and emails personalized offer letters to those parents[45]. Each email includes the acceptance deadline (as set in school settings) and a unique link to accept/decline. The notification date must appear on these emails (e.g. "You will be notified by October 5" as required by law)[46].
- **Waiting List Notifications:** Similarly, admins can email those on the waiting list (e.g. with an update or "still waiting" notice). In the demo, after sending offers, the "Send Email to Waiting List" button became enabled, and clicking it emailed the 18 waiting-list parents[45][47].
- **Timestamps:** The system logs the offer issuance with timestamp. Each offer record shows when it was sent and the response deadline. All such actions (offer sent, emails sent) must be captured in the audit trail.
- **Failure Safeguards:** To prevent mistakes, the system should disable the offer email buttons until the correct dates/times are reached (the demo admin noted "once the notification date arrives, I can send offers; before that the button is greyed out to prevent accidental sending"[45]).

11. Parent Offer Acceptance & Decline Flow

After offers are sent, parents interact with the system as follows:

- **Interface:** In the parent portal, accepted applicants see an “Offered” status and can click to view the offer letter. They have clear **Accept** and **Decline** buttons on the offer page. The demo parent accessed the link, logged in, and saw “Accept or Decline the offer that’s been sent to me”[48].
- **Acceptance:** When a parent clicks **Accept**, the application status changes to Accepted. The parent is then invited to begin the Enrollment Form (see Section 12). The dashboard changes to show one accepted but “Not started”[49]. Administrators immediately see on their dashboard that the child has accepted (status updated) but enrollment is pending.
- **Decline:** If the parent clicks **Decline**, the system marks the application as declined and frees up that seat. An email notification is sent to the school administrators stating the decline[50]. In the demo, as soon as one offer was declined, the system’s status updated to reflect that (and the admin received an email). The admin can then issue an offer to the next eligible applicant on the waiting list (see Section 15). (Alternatively, the admin could manually change someone’s status to offer via the UI; a “Bulk Promote” button could help with multiple declines.)
- **Email Notifications:** Upon decline, the parent receives a confirmation email (“Your offer has been declined”[51]) and the school gets an alert email. Similarly, if the parent accepts, they get an acceptance confirmation (in demo, the system informed the parent by email to proceed with enrollment).
- **Reminders:** The system provides reminder emails. Schools can send automated reminders to any parent who has been offered a place but not yet responded[52]. Likewise, once the waiting list is created, the school can re-email the current waiting list (for example, if priorities shift) to let them know their new rank[52]. These reminders ensure parents take timely action.
- **Seat Update Logic:** When a parent declines, or an accepted student later withdraws, the system should promote the top waiting-list applicant into an offer status. In the demo, the admin manually selected the next student (Missy Cooper) and clicked an action to send her an offer[53]. The system must support this promotion: either through a “Next on Waiting List” button or by changing the applicant’s status to “Offered” and sending the standard offer email. The waiting list order (see Section 15) must be consulted to make this fair.

12. Enrollment Form Requirements

Once an offer is accepted, the student moves into the Enrollment stage. The parent (or guardian) must complete a second, more detailed form to finalize admission.

Requirements:

- **Initiation:** After accepting an offer, the parent’s dashboard shows the option to begin the Enrollment Form. Until they click “Begin,” their enrollment status remains

“Accepted (Not Started)”[49]. Only after they click begin does the status change to “Draft,” which the school can see on its admin list[54]. (This lets admins track who has yet to start enrollment.)

- **Data Collection:** The Enrollment form covers information beyond the application, typically including:
- **Child Medical Information:** Health history, doctor’s details, health card number, allergies, etc.
- **Custody and Consent:** Questions about any custody arrangements, court orders, emergency contacts, photo permission, etc. For example, one question in the demo asked “Are there any custody arrangements in place?” If Yes, a follow-up requested details and a custody document upload[55]. The parent answered and uploaded a file, which the system stored (though a UI bug was noted where the upload indicator remained red until clicking outside[56]). The system must correctly handle such conditional questions and file uploads.
- **Student Photo Upload:** If enabled, parents upload a current photo of the child.
- **Previous School Report:** Upload the current primary school’s final report card. (In the demo, the prompt read “copy of report” and the admin noted it should say “report from current primary school.” Parents need a clear label for each required document.)
- **Photo Permissions:** A checkbox or consent: for example, “do you consent to the school using photos of your child?” The parent can agree or not[57].
- **Emergency Contacts:** Details of persons (other than parents) who may collect the child.
- **Doctor’s Details:** Name and address of family doctor.
- **Medical History:** Standard medical form questions (the system already has a built-in medical form, requiring yes/no and details for conditions). In the demo, this section worked well and required full completion[58].
- **Educational Needs:** Questions about special educational needs. If the parent indicates Yes to any need, the form asks for specifics and possibly an upload (e.g. psychological report)[59].
- **Language and Ethnicity:** Government-required demographic questions (e.g. mother tongue, ethnicity). The demo had a bug where “Is English/Irish the mother tongue?” defaulted to Yes[60]; it should be blank. The admin will need to ensure these dropdowns default to blank.
- **Anti-Bullying Policy and Others:** Final checkboxes for consents (e.g. agree to anti-bullying policy[61]). Parents should be able to view linked policy documents (the demo had “link to anti-bullying policy?” but it was missing)[61].
- **Validation:** Like the application form, all required fields must be filled before submission. The UI should clearly indicate missing info. In the demo, after

uploading the custody document and signatures, the field still showed an error until clicking outside[56] – such UI glitches must be fixed.

- **Submission:** Once complete, the parent submits the enrollment form. An email is sent confirming receipt (as happened in the demo). A combined PDF of the application + enrollment details should be generated. The demo showed that the downloaded “form” was actually still the application form[43]. The final system must provide the *enrollment form* PDF (which the admin saw could be downloaded from the applicant’s record[62]). After submitting, the parent should not be able to cancel or delete their enrollment (the school retains that authority)[63].
- **Admin Visibility:** The administrator sees the student now listed under “Enrollments.” On their dashboard and enrollment list, the student status reads “Submitted”[62]. The admin can view all entered enrollment data under the student’s record.

13. Criteria Configuration Requirements

Administrators must be able to configure admissions criteria in a flexible, ordered manner:

- **Yes/No Questions:** For each school, admins create a list of boolean questions (e.g. “Is child a sibling of a current student?”). Each question has a display text and can include instructional helper text. In the settings demo, creating a new criterion involved entering the question text and saving[64].
- **Priority (Ranking):** Criteria are inherently ranked by order. The first question has highest priority, the next lower, and so on. In the selection process, all “Yes” answers to the top-ranked criteria get offers first[40]. The system does *not* assign numeric scores; it simply follows the order. (In the demo, the admin explicitly said “There’s no scoring. Don’t do that.”[30].) The interface should allow admins to reorder criteria.
- **Follow-up Details:** For many criteria, a “Yes” answer triggers additional details. The admin must be able to specify if further information is required, and the system will then ask the parent for it. For example, if sibling = Yes, the admin enabled “Further details” and specified “Name of sibling and current class”[65]. The parent then saw fields to enter exactly those details. Up to two follow-up questions can be configured per criterion (the system supports two pieces of further information)[66].
- **Document Requests:** Admins can also demand uploads for certain answers. For instance, if a criterion has “Yes” and requires proof, the admin toggles “Require upload” for that question. The system then shows an upload field to the parent. In the admin UI, specifying this was done by checking “Document upload” for the criterion (e.g. “If Yes, please upload proof of catchment...”[67]). Note: the current UI does not allow entering explanatory text in the upload prompt, so admins often include instructions in the question text itself.

- **Dropdown Options:** Some questions may use dropdown selections instead of free text. The admin can define a criterion whose answer type is a dropdown list. The demo created a “feeder school” question as a dropdown, but found that the list of values was not visible to the parent (it appeared as static text)[68][69]. This is a known issue: the parent interface must present real clickable dropdowns. (The transcript noted that this will be fixed: “parent must be able to see what is in that list before they say yes”[68].)
- **Application vs. Enrollment Stage:** Some criteria only apply at acceptance. The admin settings allow toggling whether a question appears during application or enrollment. For instance, permission to contact primary school or upload current school report would be set to Enrollment stage. The demo’s “Optional Questions” tab showed which fields were enabled at each stage[7]. Every configurable field (gender, tongue, etc.) should have an on/off toggle and stage selector.
- **Criteria Settings Flow:** In the admin console, setting up one criterion (like “ASD class”) requires visiting multiple tabs. For example, enabling the ASD question is under “Optional Questions,” then defining its follow-ups and required documents is under “ASD Document Info”[17]. The new system should streamline this (all settings for one question should ideally be on one page).

14. Lottery & Randomization Requirements

When a criterion group has more qualifiers than remaining seats, the system runs a lottery:

- **Trigger Lottery:** During selection, the system fills seats criterion by criterion. If it reaches a criterion where qualifiers exceed seats (e.g. 11 applicants for 4 seats), it must randomly select which 4 receive offers[41].
- **Unique Random Numbers:** For transparency, the system assigns each applicant in that pool a unique random “lottery number.” These numbers cannot repeat[70]. The admin can print or export a list of these lottery tickets (the demo exported a set of numbered tickets like 12311, 12312, ... for drawing)[70].
- **Public Drawing:** The actual draw is done externally (e.g. pulling tickets from a hat) to ensure fairness. The admin or school officials then click the drawn numbers in sequence in the interface (e.g. “7476 – Confirm”[71]). Each confirmation assigns a place to that applicant. The interface shows the drawn number being awarded a seat[71].
- **Auditable Process:** The system logs each drawn number and timestamp. Once the lottery is complete (all seats assigned), the admin finalizes the process. The demo required clicking “Complete Lottery” and then moved to waiting-list generation. All lottery actions should be time-stamped in the audit. The random numbers and selections should be included in exports and applicant records for full transparency.

- **UI Considerations:** The lottery selection interface must display all candidates in the pool. The demo noted a serious issue: only 50 per page were visible, making it hard to find numbers beyond page 1[13]. Sorting and search must reliably find any ticket number. Also, after exporting the ticket list, the interface should automatically show the correct lottery numbers (the demo eventually saw them populate)[72].

15. Waiting List Management Requirements

Once all places are offered or filled, remaining applicants are ordered on a waiting list:

- **Automatic Generation:** After lottery completion and seat assignment, the system prompts the admin to create the waiting list[73]. All not-yet-placed applicants (from the oversubscribed criterion and all lower-priority criteria) are included[74].
- **Waiting List Order:** The waiting list is ordered first by admissions criteria and then by lottery ranking. All applicants from the first criterion group that wasn't fully seated go into the waiting list (preserving their relative lottery order); then all from criteria 3, 4, etc., in turn[74]. (In practice, anyone with a higher-ranked "Yes" should be above those with a lower-ranked "Yes"[75].) The interface lists everyone eligible.
- **Assigning Positions:** Administrators assign position numbers. Typically, the random-draw continues: after finishing seats, the lottery drawing can proceed down the list to rank waiting positions. In the demo, admins called out "number one is 9616" and entered it as position #1 on the waiting list, then #2 as 6844, etc[76]. The system must allow entering each position (or automatically continue the draw). Ideally, a "Save" button should let admins save partial lists; the demo noted the need to pause and continue when the list is long[77].
- **Columns – "Waiting List Number" vs "Position":** The admin interface distinguishes the lottery ticket number and the current position. In [18], the trainer explained that one column is the assigned "waiting list number" and another is the updated "position"[78]. It appears "Waiting List Number" may be the original random number or rank, and "Position" is the rank order (1, 2, 3...). The UI should clarify this labeling.
- **Parent View:** Parents should not see their waiting-list status until offered or until the school sends a waiting-list email. In the demo, even after listing positions, the parent dashboard still showed "Applied" (not "Waiting") until admin finalized and emailed them[79]. Once notifications are sent, the parent's status updates to "Waiting" on their dashboard.
- **Promotion from Waitlist:** When a seat frees up (decline or withdrawal), the system must promote the top-waiting applicant. The admin can do this manually by selecting them and clicking "Send Offer." In [21], after a decline, the admin selected the #1 waiting student (Missy Cooper) and issued her an offer[53]. The

system should also mark her status as Offered and send the standard offer email. If multiple offers open, a bulk-promote button could be used (the demo had a “Select #” button for multi-select)[80].

- **Updates:** If the waiting list order changes (e.g. by re-running a lottery or policy change), the system can send updated positions via email. The demo included the ability to resend waiting list emails, which would update everyone’s “Position” field in their email[52]. Administrators should be able to send these periodically.
- **Long Lists:** Large schools can have hundreds on the waiting list. The system should handle this gracefully. Admins should be able to paginate, search, and export the full waiting list. The interface should not lose data if partially saved (the demo admin insisted on the ability to “save” and re-enter[77]).

16. Administrator Workflow Requirements

Administrators must have a user-friendly interface and tools for all admissions tasks:

- **Dashboard:** The admin landing page shows key metrics (total places, applications received, offers sent, accepted/declined, waiting list count). For example, the demo dashboard showed “115 of an intake and 135 applications (115 offers, 18 waiting)”[81]. These stats help admins track progress.
- **Application Review:** A master list of all submitted applications is provided, with filters and search (e.g. by application number, child name, lottery number[82]). Each row shows basic info (status, applicant name, date of birth, PPS, etc.). Admins can click an application to *View* its full details in a tabbed view[83]. Tabs include **Application Details**, **Student Profile** (if enabled), **Guardian Details**, **Criteria Responses**, **Uploaded Documents**, **Audit Trail**, etc.
- **Viewing Details:** In the Application view, admins see exactly what the parent entered. For example, all student info fields, sibling info (if any), parental info, etc.[83][84]. Guardian addresses appear on the Guardian tabs; the demo admin felt guardian 1’s address should show on the same page as the guardian’s info for convenience (currently split)[84]. Under “Criteria,” admins see each question and the parent’s answers/follow-ups[85]. Under “Documents,” admins can preview or download each uploaded file (the demo showed the uploaded utility bill and birth certificate)[42]. The “Audit” or “Trail” tab shows timestamps of every transition (e.g. “Draft at 11:11, Submitted at 11:28”)[19].
- **Editing Applications:** If a parent made an error (e.g. typos), the admin can click **Edit** on any section and correct it[85]. This reopens the application in edit mode. (However, note that the system should not allow editing after submission unless unlocking the record, since the application is considered a signed statement.) A known issue is that after ~10 minutes idle, the edit buttons disappeared (soft-

logout)[18]. The system must maintain session or provide a warning so admins can continue without losing place.

- **Offers & Waiting List:** From the application list, admins can switch views or tabs to see *Offers* and *Waiting List*. Once selection is run, applications move from “Applied” to “Offered” or “Waiting” statuses. Admins can bulk-send offers to selected applications, or revoke/decline offers if needed. Similarly, the Waiting List screen allows viewing/ranking of waitlist students and sending offers to those promoted.
- **Exporting Data:** As noted, the admin can export the full dataset. The “Export to Excel” button generates CSV/Excel files with all application data, criteria answers, lottery numbers, waiting positions, and enrollment info[12]. The demo highlighted that exports come as multiple split files due to row limits; this must be fixed so a single file (or logical grouping) is generated. The columns should include meaningful headers (e.g. “Lottery Number,” “Waiting Position”).
- **Communications:** In the admin portal, there are areas to configure and send emails. Admins can preview and edit any email template (submission confirmation, offer letter, reminder, etc.) with placeholders (student name, dates, etc.)[86]. After the system sends communications (for example, after clicking “Send Offers”), admins can view a log of sent emails or even re-send if necessary.
- **Configuration Tools:** Beyond criteria, admins use the **Settings** menu (see Section 14) to adjust parameters like enrollment deadlines, consent texts, and form fields. All changes should be immediately saved (note that some demo bugs required double-saving for times[38]; this must be robust).
- **Audit Trail:** In every admin action (changing criteria, sending emails, editing applications), the system logs who did what and when. This is visible in the application’s audit tab[19]. Reports and export logs also serve for auditing.

17. Super Admin Configuration Requirements

Super Administrators (system-wide admins) have additional capabilities:

- **School Setup:** They can create and configure new schools for the system (e.g. adding a new school profile and its initial settings)[87]. This includes setting default templates and menus.
- **Global Data:** They maintain universal dropdown lists used across schools (e.g. country list, ethnic categories, medical lists). This ensures consistency. If a new category is needed (e.g. a new country code), super admins add it in one place.
- **Role Management:** They define admin roles (e.g. “School Admin,” “Document Reviewer,” “Super Admin”) and assign permissions. For example, school admins can manage only their school’s config and data.

- **System Policies:** They ensure system compliance (e.g. they may lock certain settings in line with new regulations).
- **Central Dashboard:** Super admins can see aggregate stats across all schools and run system-wide reports.

(Note: While the transcript focused on a single school’s workflow, the system must inherently support multiple schools with isolated settings and data as described in Section 2.)

18. Audit Trail & Compliance Requirements

Transparency and compliance are critical:

- **Comprehensive Logging:** Every action is recorded with user ID, role, timestamp, and description. This includes application submission, modifications (by parent or admin), offer issuance, acceptance/decline, document upload, lottery draws, and enrollment submission. The “Application Trail” shows these steps sequentially[19].
- **Permanent Lottery Records:** Lottery random numbers and results are permanent records. Even after selection, the system retains each applicant’s assigned lottery number[70]. These appear on exported reports.
- **Immutable Submission:** Once a parent submits an application or enrollment (with their signature), the record is immutable. Parents cannot alter it (only cancel). The system should maintain a cryptographic signature or similar to prevent tampering.
- **Regulatory Fields:** The system must ensure mandatory fields for compliance. For example, certain state or national ID fields (like PPS number) are required by law. The enrollment questions on mother tongue and ethnicity were noted as government-required (the UI should force a Yes/No to “Is English the mother tongue?” rather than defaulting to Yes)[60].
- **Data Protection:** The system must comply with data protection standards (e.g. GDPR where applicable). Access to personal data is strictly limited by role (school admins see only their applicants, etc.). All sensitive data (PII, documents) is encrypted at rest and in transit. Consent forms (data usage, anti-bullying, photo permission) are clearly presented for electronic signature and retained.
- **Auditor Access:** There should be a way for auditors or compliance officers to download a complete audit log or run report queries across actions.

19. Communication Templates & Workflow

Automated communications keep parents informed:

- **Template Management:** Admins can create and edit email templates for each type of notification. Templates include placeholders (e.g. student name, dates) that the system fills at send time[86]. For example, the Offer email uses the school’s letter of offer content; the admin can edit the subject line and body in Settings[88].

- **Automated Triggers:** Emails are triggered by events:
 - *Submission Confirmation:* Sent immediately upon application submission (includes attached PDF of application).
 - *Offer Letter:* Sent when admin clicks “Send Offers.” It contains offer details and deadline. The demo emails explicitly mention the final date for acceptance[46].
 - *Waiting List Notification:* Optionally sent when waiting list is created or updated.
 - *Decline Notices:* If a parent declines, both parent and school admin get emails (the parent’s says “you have declined the offer”[50]; the school’s says who declined).
 - *Reminders:* Configurable reminders for parents with pending offers, pending acceptances, or incomplete enrollments. The demo had reminder buttons to email all “Offered–No Response” or “Accepted–Not Started” lists[52].
 - *Enrollment Confirmation:* Sent upon enrollment form submission (with enrollment PDF).
 - *General Communications:* The system can also send announcements (e.g. “Application window opens”) if needed.
- **Placeholders and Personalization:** Templates support dynamic fields (parent name, student name, school name, dates, etc.). The demo showed variety of placeholder tags available[86].
- **Workflow Integration:** Communication features are integrated into the workflow buttons. For example, after running the selection, the admin clicks one button to “Send Offers” and the corresponding offer emails go out automatically[45]. Similarly, “Send Waiting List Email” dispatches to all waiting list members. The admin interface should provide feedback (e.g. “5 offers sent,” “18 waiting-list emails sent”) as seen in [20].

20. Reporting & Export Requirements

Administrators need flexible reporting:

- **Exportable Data:** All admissions data (applications, criteria responses, lottery numbers, waiting list positions, enrollment details) must be exportable as CSV/Excel. The export should flatten related data so each row is an application with all fields (yes/no answers, uploaded file flags, etc.). The system should combine all data into one comprehensive dataset with clear column headers.
- **Unlimited Rows:** Unlike the demo system (which split into multiple files at ~70 rows[12]), exports must handle thousands of records in a single file or an appropriately partitioned set (for performance) without arbitrary limits.
- **Consistency:** Exports should be consistent across runs. For example, if I export at 9am and again at 9:30, the datasets should match (unless changes happened).
- **Filtering:** Admins may request filtered exports (e.g. only accepted students, or one grade level). The system can allow specifying filter criteria before export.

- **Reporting Dashboard:** In addition to raw exports, an admin dashboard should offer summary statistics (totals by grade, criteria fulfillment counts, etc.). These help school leadership analyze applicant pools at a glance.
- **Audit Exports:** There should be an export of the “audit log” as well, listing all actions (useful for compliance).

21. Performance & Scalability Requirements

The system must remain performant under heavy load:

- **High Throughput:** It must support the peak load scenario (e.g. thousands of applications in a short span). The transcript cites that a school received ~6,500 submissions in two days[1]. The system should handle such a surge without timeouts or crashes.
- **Responsive UI:** Parent form submissions and navigation should respond quickly (aim for <2s response for page transitions, asynchronous saves). OTP verifications and email sends should also be fast.
- **Reliability:** During peak periods, the system should exhibit high availability (minimal downtime). Error rates (failed submissions, lost data) must be near zero.
- **Scalable Architecture:** The backend should scale (e.g. horizontal app servers, database replicas) to handle sudden load spikes (e.g. right after window opens). Session management (authentication) should be load-balanced and persistent.
- **Batch Operations:** The lottery drawing and waiting list assignment might involve large cohorts. Those screens must remain usable (no large performance delays when listing hundreds of candidates).

22. Security, Privacy & Data Protection

Protecting personal data is paramount:

- **Role-Based Access:** Access to student/parent data is strictly controlled. Parents only see their own data. School admins only see data for their school. Super admins see all. Each action in the UI checks permissions.
- **OTP Verification:** User registration requires email and SMS OTP verification[20]. This ensures contact data is validated (important for subsequent communications). The OTP system must be secure against brute force (e.g. rate-limit attempts).
- **Encryption:** All data in transit is encrypted (HTTPS). Sensitive data (e.g. national ID, medical details) is encrypted at rest. Uploaded documents are stored securely.
- **Consent Management:** The system stores parents’ consents (agreeing to terms, code of conduct, etc.) along with timestamps. Parents’ electronic signatures (drawn or uploaded image) are recorded.
- **Privacy Compliance:** The design should follow relevant privacy laws (GDPR, FERPA, etc.). Parents should have a way to view and, if needed, correct their data.

Document retention policies (how long to keep data after admission or withdrawal) must be implemented.

23. Data Model Narrative

The database will contain records for each application and enrollment, linked to student and guardians:

- **Application Record:** Includes student info (name, DOB, PPS, etc.), guardian info, contact preferences, submitted date, status, and any criteria answers (yes/no) and follow-ups. It also stores links to uploaded files (birth cert, etc.) and the random lottery number if used.
- **Enrollment Record:** For each accepted student, stores medical info, custody details, consents, additional contact info, etc., and links to enrollment-stage uploads.
- **Criteria Data:** Definitions of each criterion (text, priority order, follow-up questions, type of answer, whether a document is required) are stored to evaluate each application.
- **Lottery and Waiting List:** Each application's lottery number and waiting list position (if any) are saved. Lottery events and waiting list assignments are timestamped in the audit.
- **Audit Log:** A separate log table records every action (user, action type, timestamp, related application ID).
- **Communications Log:** Emails sent (to whom, when, which template) may also be logged for tracking.

The model must support historical snapshots: if an application is edited or withdrawn, a record of the prior state is kept for auditing.

24. User Experience & Accessibility Requirements

The interface must be clear, intuitive, and accessible:

- **Guided Navigation:** Parents should see progress indicators (e.g. "Step 3 of 7") so they know where they are in the form. All instructions and labels must be easy to understand.
- **Responsive Design:** The application forms and dashboards must work well on desktops, tablets, and smartphones. The transcript demo was on desktop, but many parents may use phones. All pages should reflow appropriately.
- **Keyboard Navigation & Screen Readers:** The UI should follow accessibility standards (WCAG 2.1). For example, all form fields have labels, error messages are screen-reader friendly, and the interface can be used without a mouse.
- **Error Feedback:** If the user misses a required field, a clear red highlight and message should indicate which field is missing. The admin noted that the "Upload"

field stayed in error state until clicked outside[56] – in production, form validation should be instantaneous and actionable.

- **Consistent Layout:** The design should avoid splitting logically related information across tabs if possible. The demo admin disliked having guardian addresses separate from their details[84]. Consider grouping related info on single pages or enabling inline expansions.
- **Announcements:** Important messages (e.g. “Applications close on X date”) should be prominently displayed on the parent dashboard[21]. The current admin-added “hello text” is an example of this.
- **Confirmation Screens:** After major actions (submission, offer acceptance), show clear confirmation screens/emails so parents are assured their action was received (e.g. “Your enrollment has been submitted successfully”).

25. Operational Requirements

- **Error Handling:** The system should handle failures gracefully. For example, if email sending fails (SMS OTP or offer email), it should retry and notify the admin if persistent.
- **Support Tools:** Provide administrators with logs and the ability to troubleshoot issues (e.g. view the last OTP sent, re-send email tokens). The high support volume (hundreds of cache/refresh tickets were noted[4]) implies the system should minimize such cases.
- **Monitoring & Alerts:** Administrators should have access to logs of system health (uptime, error rates). There should be alerts for outages or critical failures (e.g. database down).
- **Maintenance:** A maintenance mode allows planned downtime (with a front-end notification). The transcript speakers mentioned staging vs. demo systems[89]; similarly, a separate sandbox or staging environment is needed for testing new configurations.
- **Backup and Recovery:** Daily backups of database and file storage, with tested restore procedures, to prevent data loss in disasters.

26. Assumptions & Constraints

- The system assumes users (parents and admins) have stable internet access.
- All schools will provide necessary legal text (privacy policy, code of conduct, etc.) in advance.
- Parents will adhere to stated deadlines; late applications beyond deadline will not disrupt the selection.
- The system must accommodate any differences in admissions policy per school (provided they fit within the framework of yes/no criteria and lottery).
- There is an assumption that each child is enrolled only once per cycle; duplicate applications for one child to the same school are disallowed.

27. Open Issues / Clarifications Needed

During requirements gathering and prototype testing, several issues arose that need final decisions:

- **Signature Upload:** Parents should be able to upload a hand-drawn or image signature at registration. The demo's requirement ("sign or draw signature") currently fails[16], so this function must be implemented.
- **Application ID Visibility:** Some schools want parents to see the application number in their dashboard (the demo discussed optionally adding it[90]). Clarify whether to show or hide this from parents.
- **"Delete" Action Behavior:** When a parent deletes an application draft vs. submitted form, what is the expected behavior? (In the demo it was unclear[35].)
- **Field Defaults:** Fix defaults on demographic questions (e.g. mother tongue should not default to "Yes"[60]). All default values should be blank or neutral unless explicitly set by parent.
- **Field Configuration UI:** Make the form field toggles more intuitive. The transcript critique noted that toggling a question (like ASD) required going to a separate tab to fully configure its text and uploads[17]. It may be better to consolidate each question's settings into one page.
- **Search Functionality:** Ensure that *all* application search fields work. The instructor observed that in lottery mode, searching by lottery number did work, but other searches were inconsistent[82]. All list views should support searching by key identifiers.
- **Email Race Conditions:** The demo admin had to remind parents to clear cache to log in to accept offers[91]. Investigate and fix any session/cookie issues causing repeated login failures after email.

These open issues will be addressed during implementation to refine the design.

Sources: Requirements have been derived and clarified from the original BRD and the detailed meeting transcript[20][6][15][34][38][92][41][50][55][43]. Each functional description above is informed by the walkthrough and notes in the transcript.
